

## Supplemental Materials

Table S1. Summary statistics of the training datasets and standardized gene annotations.

| Metric                               | Bacteriophage | Prokaryote  |
|--------------------------------------|---------------|-------------|
| Total genomes                        | 16,479        | 33,469      |
| Minimum genome length                | 1.79 kb       | 207.81 kb   |
| Median genome length                 | 44.83 kb      | 4.3 Mb      |
| Maximum genome length                | 734.31 kb     | 14.2 Mb     |
| Total genera                         | 1,397         | 4,900       |
| Total proteins                       | 1,473,732     | 142,978,349 |
| Median proteins per genome           | 66            | 3,947       |
| Hypothetical / dark proteins (%)     | 60.2          | 12.7        |
| PHROG annotation (%)                 | 73.7          | -           |
| Pfam annotation (%)                  | -             | 90.5        |
| COG annotation (%)                   | -             | 92.6        |
| eggNOG OG annotation (%)             | -             | 90.3        |
| Minimum proteins per vocabulary term | 50            | 50          |
| Median proteins per vocabulary term  | 270           | 1,029       |
| Maximum proteins per vocabulary term | 19,930        | 2,069,836   |
| Standardized vocabulary size         | 582           | 11,999      |
| OTHER token (%)                      | 0.82          | 0.106       |

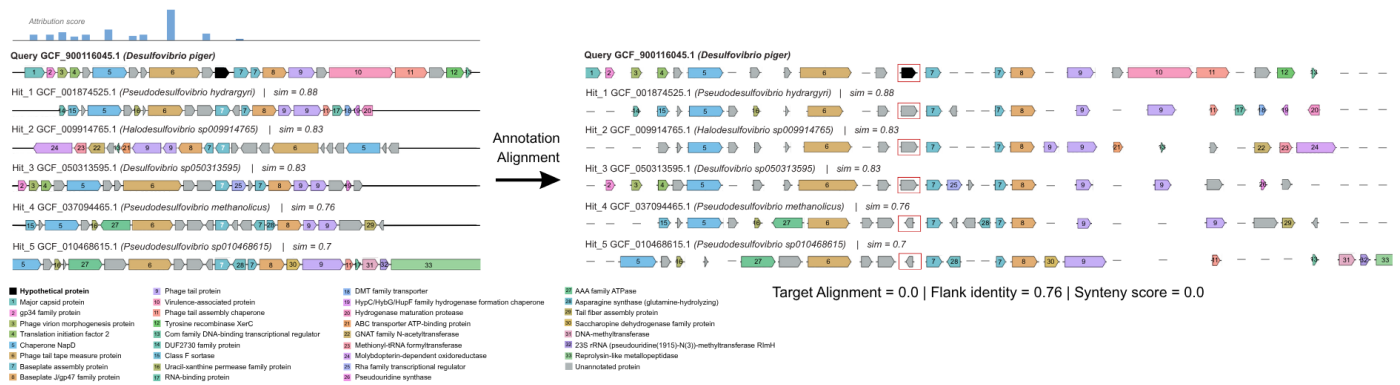


Figure S1. Example of post-hoc annotation alignment of a “context-leakage” prediction. The left panel shows neighbor attribution along with annotations of top 5 retrieved context neighborhoods pre annotation alignment, with the target gene predicted to be “Baseplate assembly protein”. The right panel shows post-alignment results, with the target gene aligned with unannotated genes in the retrieved examples, resulting in low target alignment and synteny score.

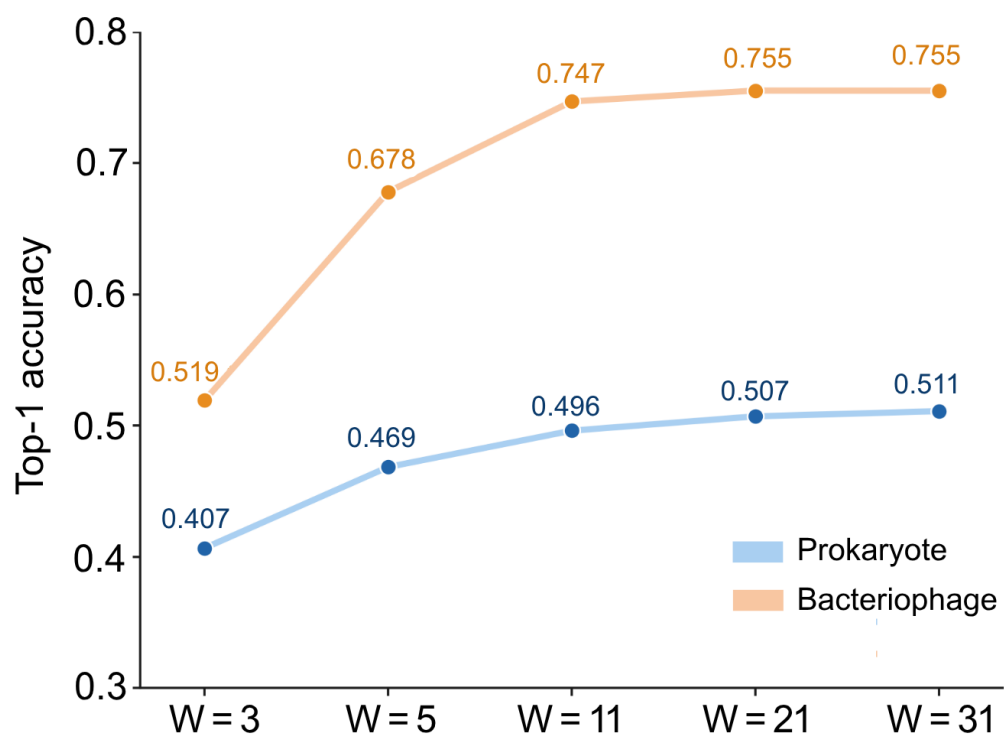


Figure S2. Synnotate performance as a function of context window length

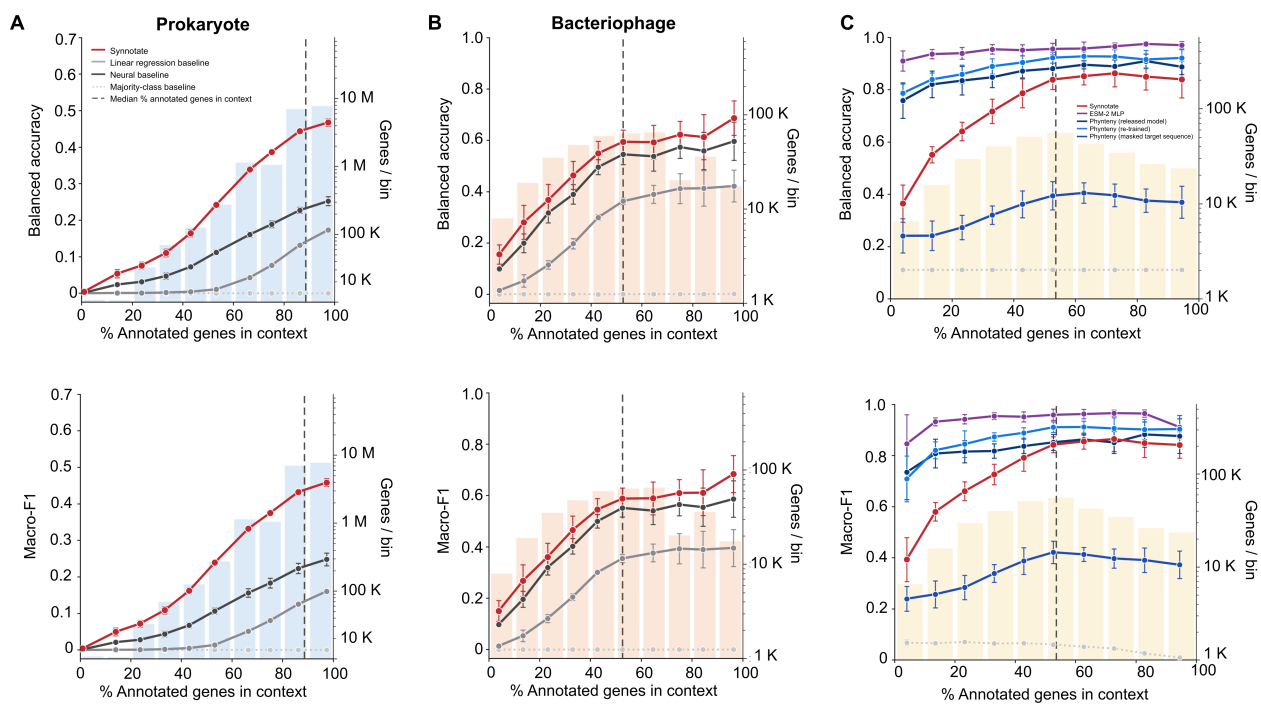


Figure S3. (A) Balanced accuracy and Macro-F1 of Synnotate on held-out genes in prokaryote datasets, stratified by the percentage of annotated genes in the 21-gene neighborhood (B) Balanced accuracy and Macro-F1 of Synnotate on held-out genes in bacteriophage datasets, stratified by the percentage of annotated genes in the 21-gene neighborhood (C) Balanced accuracy and Macro-F1 of Synnotate and Phyteny on held-out genes (9 PHROG categories) in bacteriophage datasets, stratified by the percentage of annotated genes in the 21-gene neighborhood

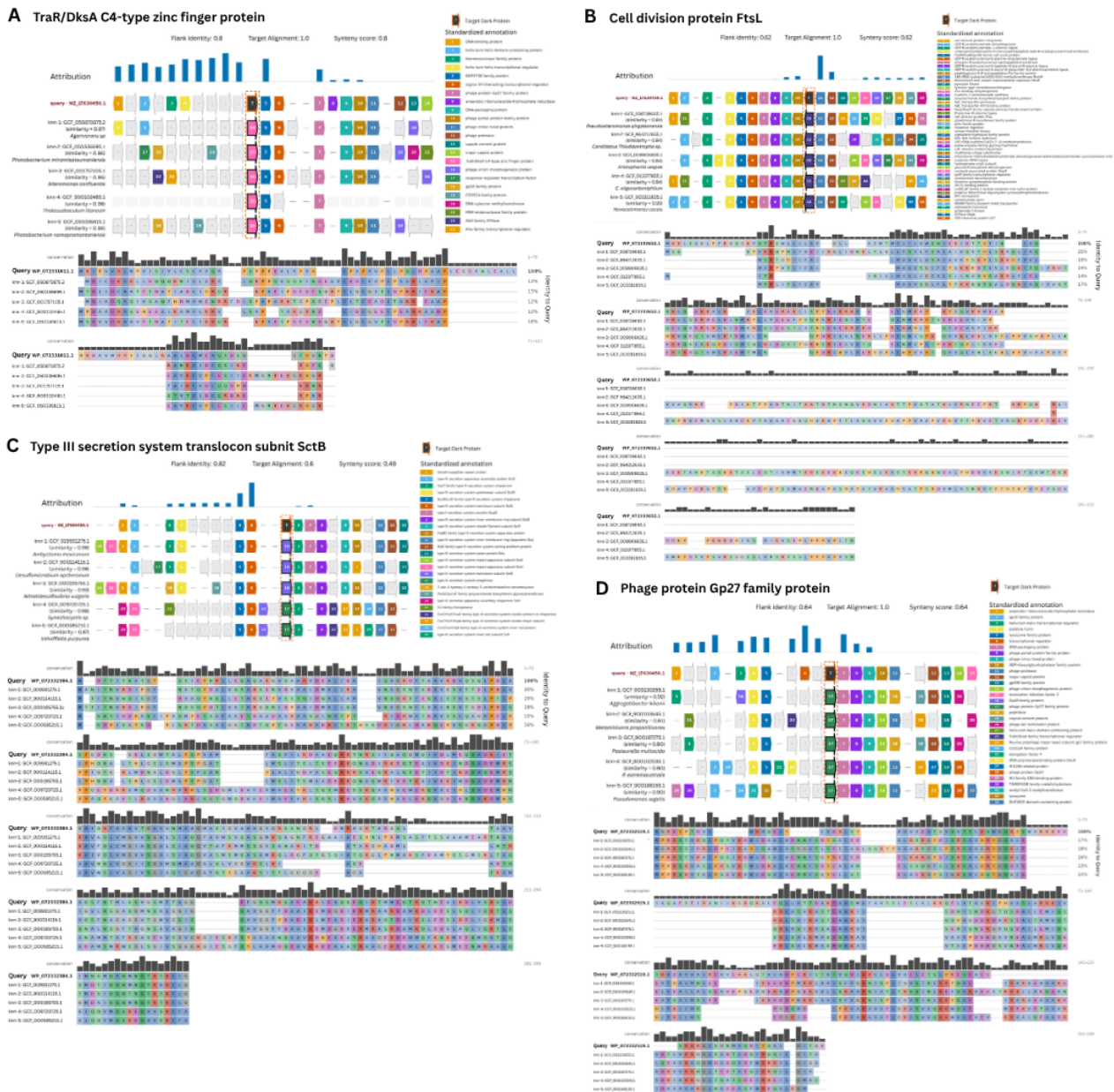


Figure S4. Multiple sequence alignment and Interpretation panels of Synnotate-annotated genes. (A) TraR/DksA C4-type zinc finger protein. (B) Cell division protein FtsL. (C) Type III secretion system translocon subunit SctB. (D) Phage protein Gp27 family protein. Protein multiple sequence alignments were generated using MAFFT<sup>33</sup> and visualized with pyMSAviz<sup>34</sup>.

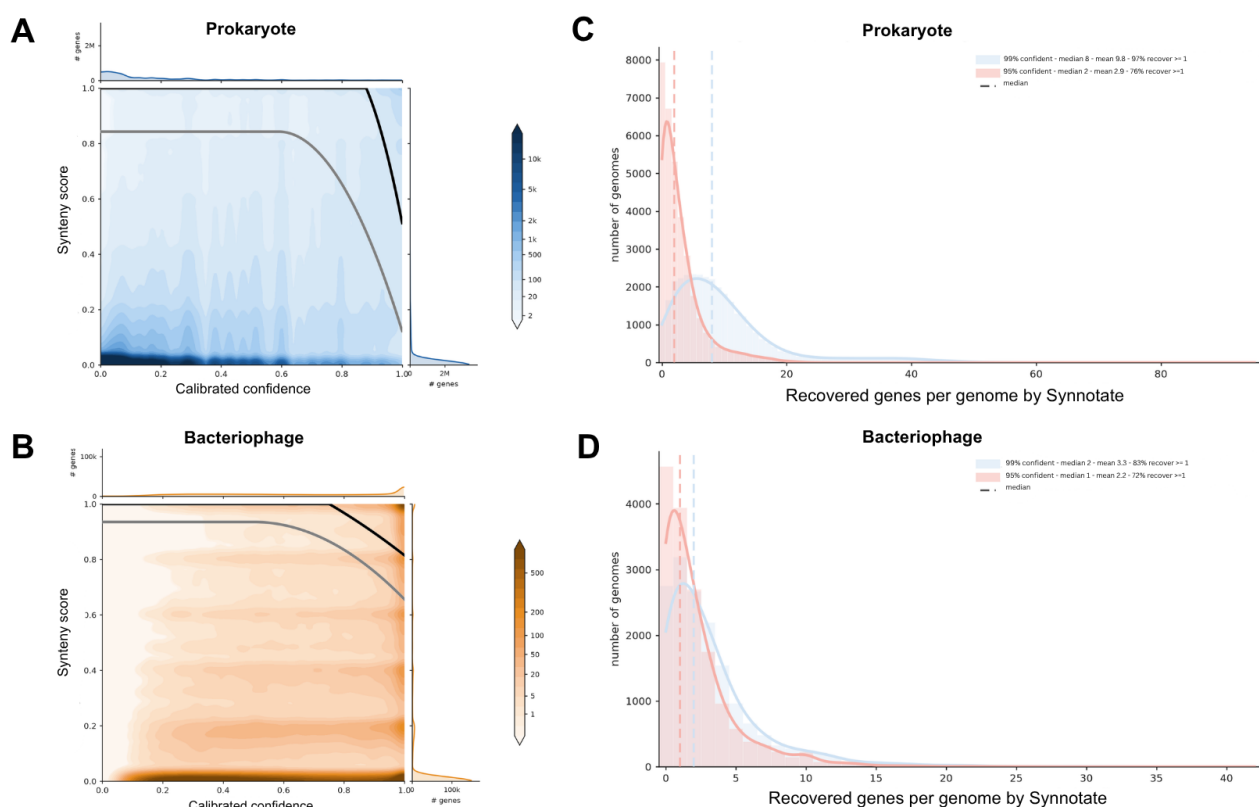


Figure S5. Annotation of the dark proteome of prokaryotic and bacteriophage datasets. (A) Synnotate calibrated confidence and Synteny score of dark genes in Prokaryote dataset. (B) Synnotate calibrated confidence and Synteny score of dark genes in Bacteriophage dataset. (C) Number of genes confidently annotated by Synnotate per genome in Prokaryote dataset. (D) Number of genes confidently annotated by Synnotate per genome in Bacteriophage dataset.

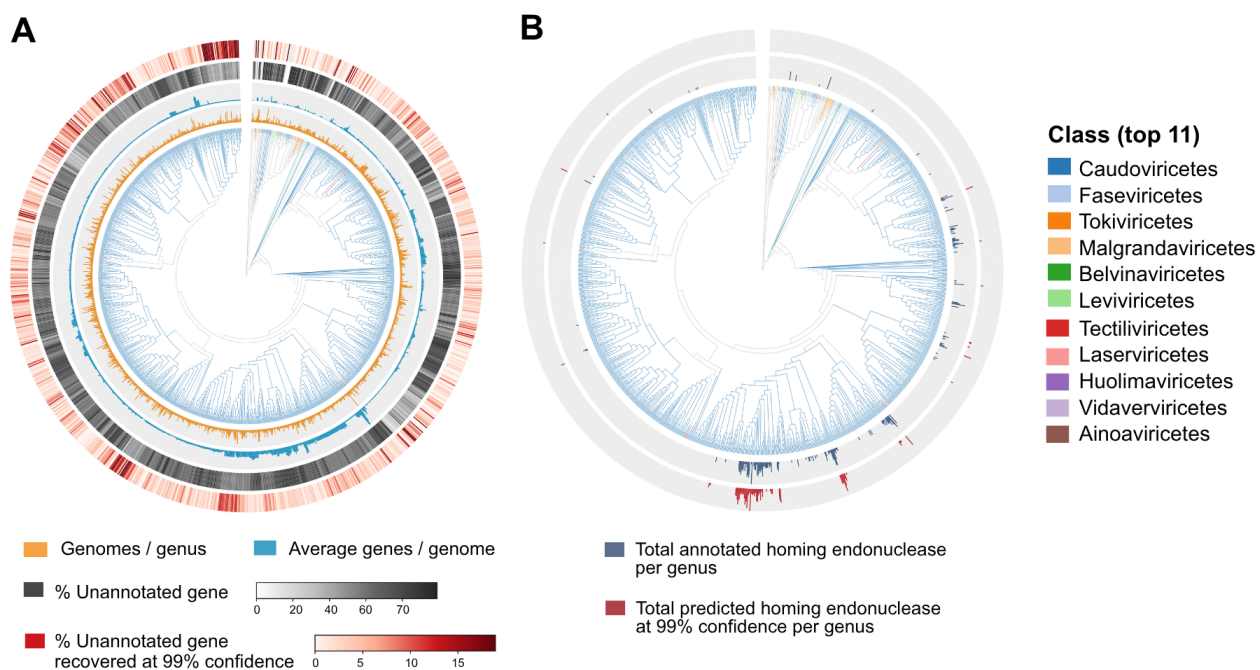


Figure S6. (A) Distribution of dark proteome and Synnotate-annotated genes across viral genera. (B) Phylogeny of annotated and predicted homing endonucleases across phage genera; each tip is aggregated at the genus level.

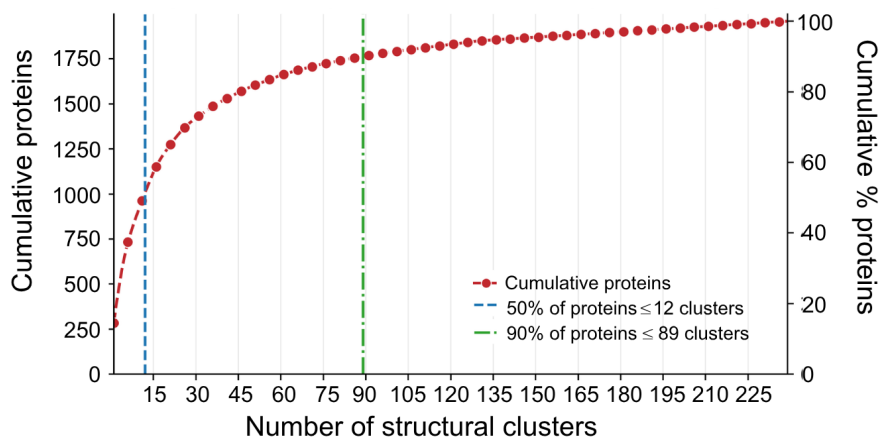


Figure S7. Cumulative distribution of structural clusters among predicted homing endonucleases